Chaos in Continuous Systems

You have already seen chaos in an iterative map called quadratic map. Most practical situations where we see chaos are however involve quantities that vary continuously. In order to mathematically model such systems we have to use differential equations. Since you have now learned how to solve differential equations numerically, using Scilab, you can now investigate continuous systems that show chaos.

Flows and Chaos

A geometric formulation of the theory of differential equations says that *a differential equation is a vector field on a manifold*. To understand this definition we present an informal description of a manifold and a vector field.

A manifold is any smooth geometric space line, surface, solid. The smoothness condition ensures that the manifold cannot have any sharp edges. An example of a one-dimensional manifold is an infinite straight line. A different one-dimensional manifold is a circle. Examples of two-dimensional manifolds are the surface of an infinite cylinder, the surface of a sphere, the surface of a torus, and the unbounded real plane (figure 1). Figure 2 shows objects whose surfaces are not manifolds. Three-dimensional manifolds are harder to visualize. The simplest example of a three-dimensional manifold is unbounded three- space, R^2. The surface of a cone is an example of a two-dimensional surface that is not a manifold. At the apex of the cone is a sharp point, which violates the smoothness condition for a manifold. Manifolds are useful geometric objects because the smoothness condition ensures that a local coordinate system can be erected at each and every point on the manifold.

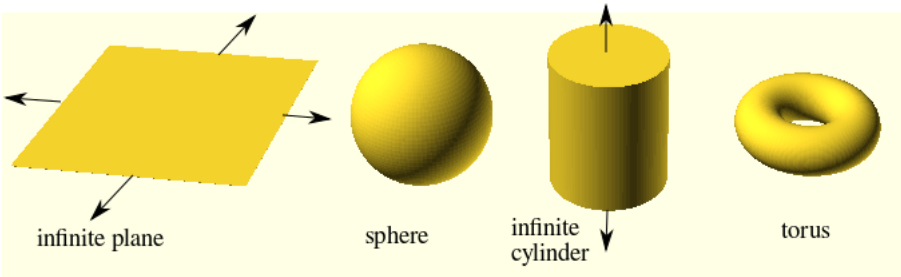

Figure 1: Surfaces of three-dimensional objects which are two- dimensional manifolds.

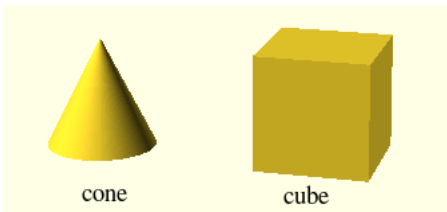

Figure 2: Examples of objects that are not manifolds.

A *vector field* is a rule that smoothly assigns a vector (a directed line segment) to each point of a manifold. This rule is often written as a system of first-order differential equations. To see how this works, consider again the linear differential equation

$$\frac{d^2x}{dt^2} = -x$$

Let us rewrite this second-order differential equation as a system of two first-order differential equations by introducing the new variable v (velocity), defined by $dx/dt = v$, so that

$$\begin{aligned} \frac{dx}{dt} &= v \\ \frac{dv}{dt} &= -x \end{aligned}$$

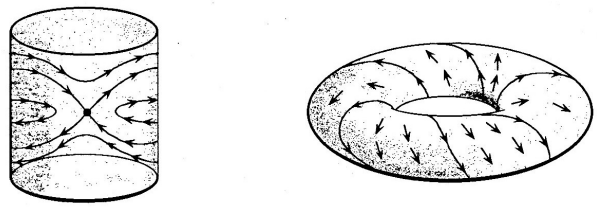

Figure 3: Examples of vector fields on different manifolds.

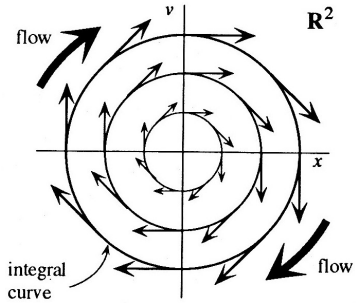

Figure 4: Vector field and flow for a linear differential equation.

The manifold in this example is the real plane, R^2, which consists of the ordered pair of variables (x, v). Each point in this plane represents an individual state, or possible initial condition, of the system. And the collection of all possible states is called the phase space of the system. A process is said to be *deterministic* if both its future and past states are uniquely determined by its present state. A process is called *semideterministic* when only the future state, but not the past, is uniquely determined by the present state. Not all physical systems are deterministic. Nevertheless, full determinism is commonly assumed in the classical scientific world view.

A system of first-order differential equations assigns to each point of the manifold a vector, thereby forming a vector field on the manifold (figure 3). In our example each point of the phase plane (x, v) gets assigned a vector $(v, -x)$, which forms rings of arrows about the origin (figure 4). A solution to a differential equation is called a *trajectory* or an *integral curve*, since it results from “integrating" the differential equations of motion. An individual vector in the vector field determines how the solution behaves locally. It tells the trajectory to “go that way." The collection of all solutions, or integral curves, is called the *flow* (figure 4).

When analyzing a system of differential equations it is important to present both the equations and the manifold on which the equations are specified. It is often possible to simplify our analysis by transferring the vector field to a different manifold, thereby changing the topology of the phase space. *Topology* is a kind of geometry which studies those properties of a space that are unchanged under a reversible continuous transformation. It is sometimes called rubber sheet geometry. A basketball and a football are identical to a topologist. They are both “topological" spheres. However, a torus and a sphere are different topological spaces as you cannot push or pull a sphere into a torus without first cutting up the sphere. Topology is also defined as the study of closeness within neighborhoods. Topological spaces can be analyzed by studying which points are “close to" or “in the neighborhood of" other points. Consider the line segment between 0 and 1.

The endpoints 0 and 1 are far away; they aren’t neighbors. But if we glue the ends together to form a circle, then the endpoints become identical, and the points around 0 and 1 have a new set of neighbors.

In its grandest form, Poincare’s program to study the qualitative behavior of ordinary differential equations would require us to analyze the generic dynamics of all vector fields on all manifolds. We are nowhere near achieving this goal yet. Poincare was inspired to carry out this program by his success with the Swedish mathematician Ivar Bendixson in analyzing all typical behavior for differential equations in the plane. As illustrated in Figure 4, the Poincare-Bendixson Theorem says that typically no more than four kinds of motion are found in a planar vector field, those of a source, sink, saddle, and limit cycle. In particular, no chaotic motion is possible in time-independent planar vector fields. To get chaotic motion in a system of differential equations one needs three dimensions, that is, a vector field on a three-dimensional manifold.

The asymptotic motions ($t \to \infty$ limit sets) of a flow are characterized by four general types of behavior. In order of increasing complexity these are *equilibrium points*, *periodic solutions*, q*uasiperiodic solutions*, and *chaos*.

An *equilibrium point* of a flow is a constant, time-independent solution. The equilibrium solutions are located where the vector field vanishes. The source in Figure 4(a) is an example of an unstable equilibrium solution. Trajectories near to the source move away from the source as time goes by. The sink in Figure 4(b) is an example of a stable equilibrium solution. Trajectories near the sink tend toward it as time goes by.

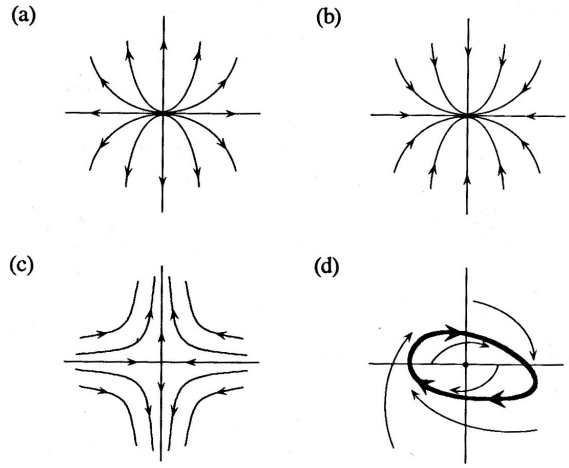

Figure 5: Typical motions in a planar vector field: (a) source, (b) sink, (c) saddle, and (d) limit cycle.

A *periodic solution* of a flow is a time-dependent trajectory that precisely returns to itself in a time T , called the *period*. A periodic trajectory is a closed curve. Like an equilibrium point, a periodic trajectory can be stable or unstable, depending on whether nearby trajectories tend toward or away from the periodic cycle. One illustration of a stable periodic trajectory is the limit cycle shown in figure 5(d). A *quasiperiodic solution* is one formed from the sum of periodic solutions with incommensurate periods. Two periods are incommensurate if their ratio is irrational. The ability to create and control periodic and quasiperiodic cycles is essential to modern society: clocks, electronic oscillators, pacemakers, and so on.

An asymptotic motion that is not an equilibrium point, periodic, or quasiperiodic is often called *chaotic*. This catchall use of the term chaos is not very specific, but it is practical. Additionally, we require that a chaotic motion is a bounded asymptotic solution that possesses sensitive dependence on initial conditions: two trajectories that begin arbitrarily close to one another on the chaotic limit set start to diverge so quickly that they become, for all practical purposes, uncorrelated. Simply put, a chaotic system is a deterministic system that exhibits random (uncorrelated) behavior.

All of the stable asymptotic motions (or limit sets) just described (e.g. sinks, stable limit cycles) are examples of *attractors*. The unstable limit sets (e.g., sources) are examples of *repellers*. The term *strange attractor* (strange repeller) is used to describe attracting (repelling) limit sets that are chaotic.

(An experimental approach to Nonlinear Dynamics and Chaos, N. B. Tufillaro, T. Abbot and J. Reilly, Addison-Wesley, 1991.)

Lorenz System

After discovering chaos in his mathematical model for weather, Lorenz simplified his differential equations until he got a set of very simple differential equations that still show chaos. This system of differential equations, known as Lorenz system, is shown below.

$$\begin{aligned}
\dot{x}_1 &= \sigma(x_2 - x_1) \\
\dot{x}_2 &= x_1(\rho - x_3) \\
\dot{x}_3 &= x_1 x_2 - \beta x_3
\end{aligned}$$

This contains three parameters ρ, σ and β. For certain values of these parameters Lorenz system shows periodic behaviour and for some other values it behaves chaotically. A few trajectories of the Lorenz system are shown in figure 6.

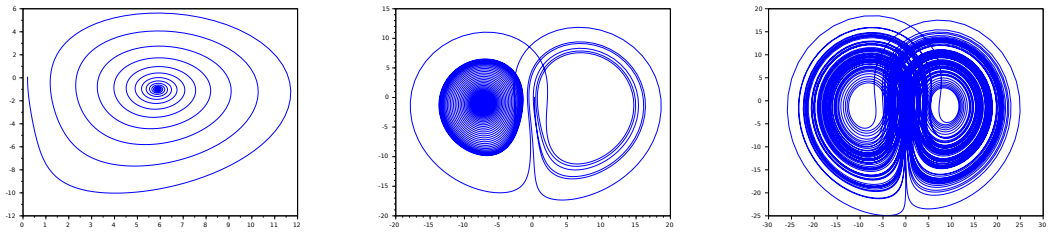

Figure 6: Phase space trajectories of the Lorenz system.